GENERAL CONCEPTS IN EPIDEMIOLOGY OF COMMUNICABLE DISEASES

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INTENDED LEARNING OUT COME

- **Define infectious, contagious, communicable disease .**
- **Define common terms used in field of epidemiology .**
- **Identify components of infection cycle.**

CONCEPT OF EPIDEMIOLOGY

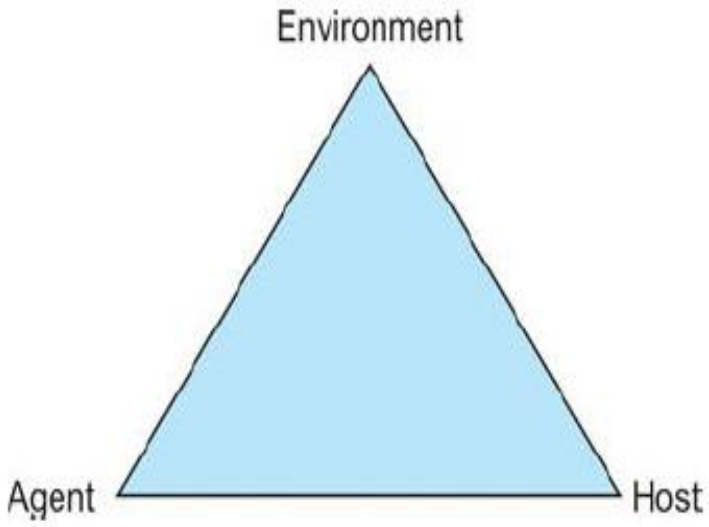
- **Epidemiology is a scientific study of factors and conditions related to disease as they occur in people.**
- **The word epidemiology is derived from epi (in, on,upon); demos (people) and logos (science).**

DEFINITION OF EPIDEMIOLOGY

- **Epidemiology has been defined as, 'The study of distribution and determinants of health-related states or events in specified populations and application of this study to the control of health problems' (John M Last, 1988).**

THE EPIDEMIOLOGICAL TRIAD:

The occurrence and manifestations of any disease, whether communicable or noncommunicable, are determined by the interactions between the agent, the host and the environment, which together constitute the epidemiological triad.



THE EPIDEMIOLOGICAL TRIAD:

Agent factors: disease agents may be classified into:

- Biological: e.g. bacteria, viruses, rickettsia
- Nutritional: e.g. increase of decrease of nutrients
- Chemical: e.g. toxins, dyes, CO
- Physical: e.g. radiation, heat
- Mechanical e.g. moving parts of machinery.

HOST FACTORS:

These may be classified into:

- Immunity and resistance
- Personal habits as smoking, drinking
- Age sex, race

ENVIRONMENTAL FACTORS:

- These may be classified into:
  - Physical as weather, geography, geology
  - Socio-economic as income, occupation, education
  - Cultural: e.g. traditions, beliefs, behaviors
  - Biological factors: micro and macro organisms

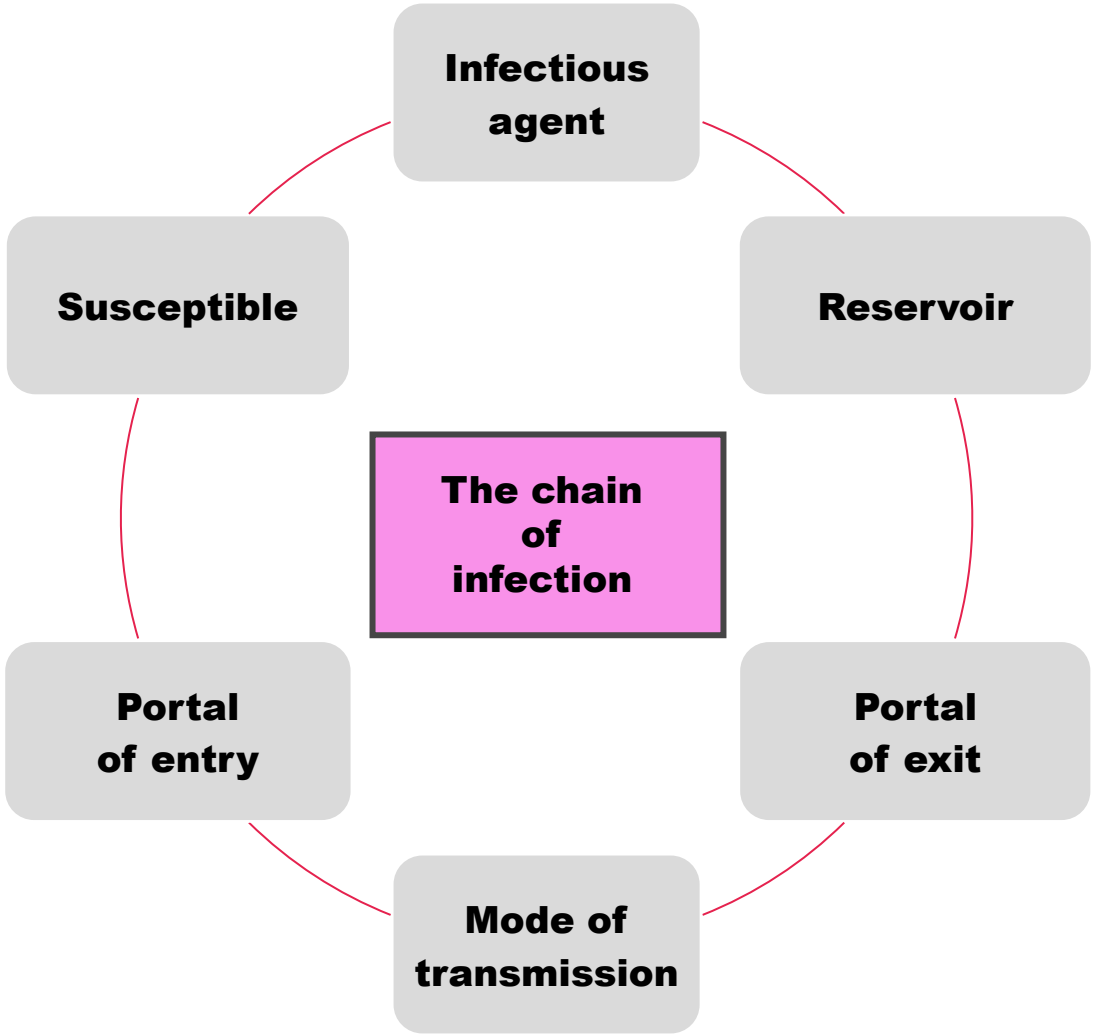
LEVEL OF DISEASE IN THE COMMUNITY

Sporadic	**Disease occurs infrequently and irregularly.**
Endemic	**Constant presence or usual prevalence of a disease in a population/geographic area.**
Epidemic	**Sudden increase in cases above what is normally expected in a population/area.**
Outbreak	**Same as epidemic, but often used for a more limited geographic area.**
Pandemic	**Epidemic that has spread across several countries or continents, affecting large numbers of people.**

IMPORTANT DEFINITIONS

- **Contamination** → Infectious organisms present on inanimate objects (clothes, utensils, food, water, blood, etc.).
- **Infection** → Entry, development, and multiplication of pathogens in a living host (may or may not cause disease).
- **Infectious Disease** → Disease directly resulting from infection.
- **Contagious Disease** → Spread person-to-person via close physical contact (e.g., scabies, STDs).
- **Communicable Disease** → Illness due to infectious agent/toxin, transmitted directly or indirectly from reservoir to host.
- **Zoonotic Disease (Zoonosis)** → Disease naturally transmitted between vertebrate animals and humans (both directions).

The chain of infection (Epidemiological or infectious cycle)



I. MICROBIOLOGIC AGENTS

The microorganisms responsible for the causation of infectious diseases are classified into

Bacteria:

- **Cocci e.g. pneumococci, meningococci, staphylococci, streptococci.**
- **Bacilli e.g. salmonella typhi and para typhi, shigella, brucella, corynebacterium diphtheria and tubercle bacilli.**
- **Spirochaetes e.g. Treponemas e.g. T. pallidum (Syphilis).**
- **Vibrios e.g. Vibrio cholara and Vibrio El- Tor.**

- **Viruses**: e.g. measles, mumps, chicken pox, influenza, common cold, poliomyelitis, small pox, yellow fever, dengue fever, and herpes zoster.
- **Rickettsiae**: Rickettsiae are usually transmitted to man by biting insects.
- **Protozoa**: e.g. Malaria & Amoebic dysentry.
- **Fungi** : e.g :Ring worm of the scalp and of the body.
- **Parasites** : e.g. Bilharziasis, Elephantiasis, and Ancylostomiasis.

RESERVOIR

- **Reservoir of Infection**
  - **Natural habitat where the causative organism lives, multiplies, and survives.**
  - **Examples: humans, animals, soil.**
- **Source of Infection**
  - **Person, animal, arthropod, soil, or substance from which the organism directly enters the host.**
  - **Reservoir ≠ Source (not always synonymous).**
- **Homologous Reservoir State**
  - **Reservoir and susceptible host are the same species.**
  - **Examples: Chickenpox, Measles, Poliomyelitis (Human → Human).**
- **Heterologous Reservoir State**
  - **Reservoir and susceptible host are different species.**
  - **Examples: Salmonellosis (Cattle → Human), Rabies (Dog → Human).**

- **Reservoirs are of three types—Human, Animal and Soil.**

- **Human Reservoir: May be case or carrier.**

Cases:

- **A case is an infected person having clinical features of the disease.**
- **Such a case may be mild, moderate or a severe case.**
- **Mild cases are more dangerous more dangerous -- (ambulatory, spread disease widely).**
- **Cases do not act as source throughout the duration of illness.**
- **Cases are infectious during the 'period of communicability', which varies from disease to disease.**

CARRIER:

- A carrier is an infected person but not having the clinical features of the disease (thus harboring the organisms) and serves as a source of infection to others in the community.
- Such carriers are seen in typhoid, diphtheria, gonorrhea, AIDS, hepatitis B, meningitis, salmonellosis and amoebiasis.

EPIDEMIOLOGICAL IMPORTANCE:

- Carriers form the "submerged iceberg" phenomenon—far more numerous than visible cases.
- They maintain endemicity of diseases in communities.
- They can spread infection to new areas otherwise free of disease.
- Their detection and treatment are essential but challenging in community medicine.

CARRIERS ARE DANGEROUS RESERVOIRS OF INFECTION BECAUSE :

- Carriers show **no clinical symptoms**, so they appear healthy.
- They and their contacts are **unaware of the risk**, so precautions are not taken.
- Carriers are **hard to detect,** often requiring laboratory tests.
- They **cannot always be managed effectively** (e.g., a typhoid carrier preparing food).
- Carriers are **more numerous** than clinical cases and **move freely** in the community, spreading infection.

CLASSIFICATION OF CARRIERS

A. ACCORDING TO THE STAGE IN THE DISEASE CYCLE (CHRONOLOGICALLY)

- **Incubatory carrier:** Spreads infection during the incubation period, before symptoms appear (e.g., measles, mumps, diphtheria, polio, pertussis, influenza, hepatitis A & B).
- **Convalescent carrier:** Spreads infection during recovery, when clinically cured but still harboring pathogens due to incomplete treatment (e.g., diphtheria, typhoid, amoebiasis).
- **Healthy/contact carrier**: Subclinically infected, never develops disease, but acts as a source of infection (e.g., diphtheria, cholera); often seen in doctors, nurses, attendants, or family members in close contact with patients.

ACCORDING TO DURATION:.

- **Temporary carriers**: Spread infection for a short period (**days**); includes incubatory, convalescent, and healthy/contact carriers.
- **Chronic carriers**: Spread infection for a long period (**weeks to months or lifelong**) due to persistence of organisms in body organs (e.g., gallbladder in typhoid, tonsils in diphtheria, liver in hepatitis B).

ACCORDING TO ROUTE OF EXIT:

- **Urinary carriers**, where the focus of organisms is the kidney as in typhoid
- **Intestinal carriers**, where the focus of organisms is the intestine, as in typhoid, amoebiasis
- **Biliary carriers**, where the focus of organisms is the gallbladder as in typhoid
- **Cutaneous carriers** as in staphylococci
- **Nasal carriers** as in nasal diphtheria
- **Genital carriers** as in gonorrhea, AIDS.

DANGEROUS GROUP OF CARRIERS

Certain occupational groups giving public services or care of vulnerable :

- **Food handlers**
- **Nurses in care of children.**
- **Hospital personnel**
- **School personnel**

Animal reservoir	Diseases transmitted
Cattle	Bovine tuberculosis, salmonellosis, Tetanus, brucellosis, Q-fever, Taenia-saginnata.
Horses	Tetanus
Dogs	Rabies, hydatid disease
Monkeys	Yellow fever, dengue fever.
Sheep	Anthrax, liver fluke (Fasciola hepatica)
Rodents	Plague, leptospirosis, endemic typhus.
Birds	Psittacosis, histoplasmosis, ornithosis.

- **Zoonotic diseases can also be classified**

1 .Viral: Rabies, yellow fever, dengue fever, Japanese encephalitis.

2 .Bacterial: Salmonellosis, brucellosis, leptospirosis, plague.

3 .Parasitic: Teniasis, hydatid disease, leishmaniasis.

4 .Rickettsial diseases:

- **a. Typhus group: Rickettsial zoonoses, epidemic typhus, scrub typhus: Murine typhus**
- **b. Spotted fever group: Tick typhus, Rocky Mountain spotted fever, Rickettsial pox**
- **c. Others: Q fever, trench fever.**

SOIL RESERVOIR

- The soil is considered the reservoir in coccidioidomycosis, histoplasomosis and probably in blastomycosis.
- Soil acts as source of infection and transmit diseases like tetanus, gas-gangrene, botulism, ankylostomiasis and anthrax.

III. PORTAL OF EXIT

- **Microorganisms must leave their reservoir (the host or environment where they live) through a portal of exit.**
  - **Natural outlets: Usually through secretions or excretions linked to the site of infection (e.g., feces, urine, respiratory droplets, skin discharges).**
  - **Unnatural outlets: Sometimes through less common routes, such as insect bites or rare practices like bloodletting.**

- **Alimentary canal**
  - **Feces: typhoid, paratyphoid, dysentery, helminth eggs**
  - **Vomitus: cholera, food poisoning**
  - **Gall bladder: enteric fever**
- **Respiratory tract**
  - **Diseases: measles, influenza, TB, diphtheria, pneumonia, plague**
  - **Spread via droplets (coughing, sneezing, talking, breathing)**
  - **Mouth = main exit; nasal secretions spread indirectly**
- **Urinary tract**
  - **Urine in systemic infections: typhoid, undulant fever**
  - **Local genito-urinary infections: gonorrhea, venereal diseases**
  - **Parasite eggs: *Schistosoma haematobium***

- **Skin & mucous membrane discharges**
  - **Skin lesions**: chickenpox, smallpox, impetigo, bacterial infections
  - **Mucous membranes**: conjunctival sac (conjunctivitis)
- **Insect bites (vector transmission)**
  - **Mosquitoes**: malaria, yellow fever, dengue, filariasis, West Nile, encephalitis
  - **Lice**: epidemic typhus, relapsing fever, trench fever
  - **Fleas**: plague, murine typhus
  - **Ticks**: relapsing fever, Rocky Mountain spotted fever, tick fever
  - **Sand flies**: kala azar, sand fly fever
  - **Tsetse flies**: trypanosomiasis
- **In-utero transmission**
  - To fetus: German measles, smallpox, syphilis (spirochetes)

IV. MODES OF TRANSMISSION

- **A communicable disease is transmitted from the reservoir/source to a susceptible person either directly or indirectly, depending upon whether they are in close proximity or not.**

DIRECT MODES OF TRANSMISSION

- There are five direct modes:
- Direct contact:
  - When the reservoir and the susceptible person are in close, physical, skin to skin contact.
  - Example, scabies, STDs AIDS, lepromatous leprosy, ophthalmia neonatorum (these are also called as contagious diseases).
- Droplet infection:
  - When the susceptible person inhales the infected droplets, of 5 to 10 μ diameter, coming out of the mouth or nose from the reservoir during the act of coughing, sneezing, talking and laughing.
  - For example, pulmonary tuberculosis, diphtheria, measles, pertussis, etc.

- **Contact with the contaminated soil:**
  - **For example, all soil borne diseases, e.g. tetanus, anthrax, ancylostomiasis.**
- **Contact with the animal:**
  - **Classical example is rabies transmitted from a rabid animal by bite and inoculating the virus into the skin or mucous membrane.**
- **Transplacental (vertical) transmission:**
  - **Means transmission of the disease from the mother to the fetus through the placenta. For example, toxoplasmosis, rubella, cytomegalovirus, herpes virus (TORCH-agents) syphilis, and AIDS.**

INDIRECT MODES OF TRANSMISSION

- There are five indirect modes:
- Vehicle route:
  - The vehicles as water, food, milk, ice-cream, vegetables, biological products like blood, tissues or organs (as in organ transplantation) can transmit disease.
- Vector route: The vectors transmit the diseases in the following ways:
  - Mechanically: e.g., house-fly transmitting typhoid, trachoma, etc.
  - Biologically: When the pathogen undergoes multiplication or development

- **biological mode of transmission is of three types:**
  - **Propagative type:** only **multiplication** inside the vector. E.g. plague bacilli in rat flea; yellow fever in female Aedes mosquito.
  - **Cyclopropagative** type: **multiplication** and **development** . E.g. malaria in female anopheline mosquito.
  - **Cyclodevelopmental type**: only development. E.g., microfilariae developing into larval stage female culex mosquito; Guineaworm embryos developing into larva in Cyclops.

- **Air-borne route of transmission:** By droplet nuclei and infected dust
  - **Droplet nuclei (<5 μm):**
    - Tiny droplets expelled from mouth or nose lose moisture, leaving pathogen residues that remain suspended in air and spread widely, causing epidemics/pandemics
    - (e.g., influenza, SARS, measles, tuberculosis).
  - **Infected dust (>10 μm):**
    - Larger droplets settle on surfaces and mix with dust; when disturbed (sweeping, bed-making, wind), they re-enter the air and spread infections in hospitals and wards
    - (e.g., pulmonary TB, streptococcal and staphylococcal infections).

- **Fomite route of transmission:**
  - **Fomites are the inanimate objects capable of transmitting the diseases.**
  - **E.g. clothes, linen (Scabies), clinical thermometer, spoons, pencils, tongue depressor (Diphtheria, streptococcal infections) surgical instruments, syringes, needles, toothbrush, shaving brush, razor (Hepatitis B, AIDS), etc. Such infections are also called as 'mediate infections.**
- **Unclean hands and fingers:**
  - **These contaminate the food and drinks and transmit diseases like typhoid, streptococcal and staphylococcal infections, diarrheas and dysentery.**

V. PORTALS OF ENTRY (INLETS)

- **Infectious agents enter the body of the susceptible through various inlets which can be classified into:**
  - **Natural inlets: Include the mouth, nose, rectum, conjunctival sac, and urethra.**
  - **Un-natural inlets: Include inoculation through blood sucking insects or through injections.**
- **Pathology depends on entry site – The portal of entry often determines the disease type:**
  - **Tuberculosis/Anthrax: respiratory entry → pulmonary form; intestinal entry → intestinal form; skin entry → dermal form.**

VI. SUSCEPTIBLE HOST

- A susceptible host is a person or animal lacking immunity to a specific pathogen, making them liable to contract disease upon exposure.
- Exposure may or may not result in infection.
  - If infection produces detectable pathology, the individual is considered to have an infectious disease.
  - Infected persons include patients, subclinical infections, and carriers.

FACTORS AFFECTING OUTCOME OF INFECTION

- Pathogenicity and virulence of the micro-organism.
- Antigenic power of the micro-organism.
- Duration of infectious state (period of communicability).
- Ease of communicability of the disease.

- **Pathogenicity & Virulence**
  - ***Pathogenicity***: ability to cause illness; higher pathogenicity → more clinical cases.
  - ***Virulence***: ability to cause severe or fatal disease; measured by case fatality rate.

- **Antigenic Power of the Microorganism**
  - Ability to stimulate immunity and antibody formation.
    - Strong antigenic power → lifelong immunity (e.g., measles, mumps, chickenpox).
    - Weak antigenic power → reinfections common (e.g., common cold, syphilis, gonorrhea).
  - Measured by second attack frequency and age-specific attack rates.

- **Period of Communicability (Duration of Infectious State)**
  - **Time during which the pathogen can be transmitted.**
  - **Short (e.g., mumps) or prolonged (e.g., typhoid carriers may remain infectious for life).**

- **Ease of Communicability of the Disease**
  - **Measured by secondary attack rate (spread among susceptible contacts during incubation period).**
  - **Indicates how easily a disease spreads within a community.**

Thank you